stedSafe

Product brief, market analysis & build specification

Calm, place-based alerts from the Norwegian police log (Politiloggen)

What this document is

A complete rundown of the stedSafe idea written from the perspective of a product manager and startup advisor, and a build specification precise enough to hand to a coding agent. It consolidates the research, the precision test of the police-log data, the competitive scan and the chosen prototype design (direction A).

How to read it

Sections 1–6 explain why and what: summary, problem, market, data findings, strategy, MVP scope.

Sections 7–9 are the normative build spec. Words in capitals (MUST, SHOULD, MAY) are requirements.

Sections 10–15 cover go-to-market, business model, risks, KPIs, roadmap and open questions.

Appendices hold reference data: full test tables, parsing rules, test fixtures, glossary, sources.

For the coding agent: start here

- Read section 8 (Architecture) and section 9 (Geocoding & classification) first; they are the core.
- The reference implementation of the geocoder is `research/presisjonstest.py` (v4). Port its logic; do not reinvent it. Appendix C lists fixtures it must pass.
- Politiloggen field names are NOT officially documented. Inspect a live response before writing parsers and keep parsing defensive (section 8.3).
- Never cache failed external lookups. Never push the same event twice to the same place.

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1. Executive summary

stedSafe turns the Norwegian police operations log (*Politiloggen*) into calm, precise alerts for the specific places a person cares about: a child's school, home, the cabin, the office. The official Politiloggen app only lets users follow a whole police district or municipality, which in Oslo means a steady stream of irrelevant traffic messages. Users have asked for map views and alerts near a location; the nearest independent competitor offers radius alerts but has a weak experience and is iOS-only.

63 %
of relevant Oslo incidents placeable within ~1 km

28 %
placeable within 500 m (street level)

~2 / day
non-traffic incidents in all of Oslo, so alerts per place are rare

0 kr
data cost: Politiloggen and Kartverket are open (NLOD)

Verdict

Worth building as a focused MVP. The demand is visible, the data is free and good enough for *neighbourhood-level* alerts, and the competitive bar is low on experience. The idea is not defensible on data (anyone can call the same API), so the advantage must come from a better, calmer and more trustworthy product, speed of execution, and Android plus iOS from day one.

What to build first

- Named places created from templates (Skole, Hjem, Hytte, Jobb, Annet), each with radius, categories and a time window.
- **1 km as the default radius**. Below 1 km, only street-level incidents count, and the app says so.
- Honest precision on every incident: *Gatenivå*, *Nabolag* or *Bydel*.
- Push **only** for serious incidents inside a place; everything else goes into a weekly digest.
- Planned traffic controls are filtered out; traffic is off by default for schools.
- Expo (iOS + Android), FastAPI, PostgreSQL + PostGIS, Expo Push. No user accounts.

Biggest risks

- **Platform risk:** the police could add radius alerts to their own app, or change the unversioned API.
- **Precision ceiling:** about 1 in 4 incidents can be placed at street level; 18 % cannot be placed at all.
- **Fear dynamics:** crime-alert apps (e.g. Citizen) are criticised for creating anxiety. Calm design is a requirement, not a style choice.
- **Privacy:** a child's school plus a home location is sensitive. Data minimisation and easy deletion are mandatory.

Next 6 weeks in one line

Interview 3–5 parents with the prototype while building the ingestion + geocoding backend; ship an internal build with push in week 4; closed beta with 20–30 Oslo users in week 6.

2. Problem & users

2.1 The problem

- Politiloggen is the police's primary channel for live incident information, but its app filters only by police district or municipality plus category. For Oslo that is far too coarse: a parent who cares about one school gets the whole city.
- Police messages carry no coordinates. Location is a free-text `area` field (e.g. «Grünerløkka», «E6 Furuset») plus street names inside the message text.
- Generic «crime near you» apps tend to maximise engagement through fear. People want awareness without anxiety.

2.2 Target users

Segment	Job to be done	Priority
Parents (school, kindergarten, after-school)	«Tell me if something serious happens near my child's school during school hours, and nothing else.»	Primary
Residents (home, rental property)	«Let me know about break-ins, fires or violence in my neighbourhood.»	Primary
Cabin owners	«I'm not there. Tell me about break-ins and fires near the cabin.»	Secondary
Commuters	«Is something blocking my route today?» (needs road data; v2)	Later
Home buyers	«What has happened in this area over the past year?» (history view; v2)	Later
Organisations (FAU, housing co-ops, cabin associations)	Shared alerts for a school or estate (B2B; later)	Later

2.3 What «good» looks like for the user

- Setting up a place takes under 30 seconds.
- Before saving, the user sees roughly how many alerts that place would have produced in the last 30 days.
- A push arrives within ~2 minutes of the police publishing, and only when it matters.
- Every incident says how precisely it is placed and why the user received it.
- A quiet week is reported as «Rolig», which is reassuring rather than empty.

3. Market & competition

Snapshot from web research on 11 November 2026. Ratings and features change; re-check before launch.

Product	Area model	Alerts	Notes
Politiloggen (Politiet, official)	Police district or municipality + category	Push per district/municipality	Free, no login. App Store NO shows ~11k ratings, 4.7. Reviews ask for a map, «alerts only near my GPS position» and English.
Politipuls (independent, solo developer)	Map, municipality pages, radius per App Store listing; up to 20 municipalities	Push by district / municipality / radius	Web + iOS, launched May 2026, 5 ratings. Adds road data from Statens vegvesen. Founder test: no visible way to set own radius, weak UI/UX. No Android app found.
Allemannsdata	Place-name search incl. sub-areas	None (developer tool)	MCP server wrapping Politiloggen for AI agents.
Politikontroller	Location-based	Push near location	Crowdsourced police checks, 500k+ Norwegian drivers. Adjacent; proves demand for location-based police info.
Poliskollen (Sweden)	Region + incident type	Push	Map, AI assistant and gamification; Pro subscription.
Brottsplatskartan (Sweden)	Map + area	Home Assistant sensor	Shows enthusiasts automate area tracking.
Nabohjælp (Denmark)	Up to two addresses (e.g. home + summer house)	Neighbour warnings	~290k users; piloting police burglary data in the app.
Citizen / Ring Neighbors (US)	GPS radius	Real-time push	Large, but widely criticised for creating a «culture of fear».

3.1 Where stedSafe wins

Dimension	Politiloggen	Politipuls	stedSafe (target)
Unit of interest	District / municipality	Municipality / radius	**Named places** with templates
Precision shown	No	Partly (location field)	**Always**, as Gatenivå / Nabolag / Bydel
Noise control	Category filter	Category filter	**Severity-gated push**, time windows, weekly digest, control filter
Platforms	iOS + Android	iOS (+ web)	**iOS + Android**
Tone	Neutral feed	Data/map	**Calm by design**

3.2 Honest assessment

- «Radius alerts from Politiloggen» alone is no longer novel. Differentiation is **experience and trust**: templates, honest precision, calm notifications.
- No data moat. Defensibility comes from habit (places set up, digest read weekly), brand trust, and later from improved geocoding and multi-source coverage.
- The police adding a radius option is the main existential risk. Mitigation: be meaningfully better (multi-place, templates, digest, precision UX) and add sources the police app will not (roads, weather warnings).

4. Data foundation & precision analysis

4.1 Sources

Source	Use	Access
Politiloggen API `api.politiloggen.politiet.no`	Incidents (threads of messages)	Open, NLOD 2.0, no key. Undocumented field names; format may change without notice.
Kartverket address API `ws.geonorge.no/adresser/v1`	Street → address points; exact addresses	Open, no key
Kartverket place-name API `ws.geonorge.no/stedsnavn/v1`	Area names → point + object type	Open, no key
Kartverket kommuneinfo API `ws.geonorge.no/kommuneinfo/v1`	Municipality name → number (needed outside Oslo)	Open, no key

4.2 Method

A research script (`presisjonstest.py`, four iterations) fetched every Politiloggen thread in the API's `LastQuarter` window ending 11 Nov 2026, kept the 424 threads in Oslo municipality, and tried to place each incident with Kartverket data. Each placement gets an **uncertainty radius**: measured for streets (90th percentile distance from the centroid of the street's address points), fixed for exact addresses (25 m), computed from the gap between the two streets for crossings, and assumed per object type for place names. 28 planned traffic controls were identified and excluded, leaving **396 incidents**.

4.3 Results (v4)

Placement method	Count	Share	Typical uncertainty
Exact address (street + house number)	4	1 %	25 m
Crossing of two streets	12	3 %	75–150 m
Street from area field	24	6 %	measured per street
Street from message text	77	19 %	measured per street
Long street limited to area	14	4 %	~500–900 m
Place name from area field	195	49 %	assumed by type (mostly 1 km)
Municipality only (not placeable)	70	18 %	n/a

Uncertainty ≤	300 m	500 m	1 000 m	2 000 m
Share of 396 incidents	21 %	28 %	63 %	75 %

Category	Incidents	≤ 500 m	≤ 1 000 m
Trafikk	195	23 %	57 %
Brann	66	44 %	77 %
Voldshendelse	42	31 %	71 %
Ro og orden	26	27 %	54 %
Andre hendelser	22	32 %	64 %
Savnet	21	5 %	62 %

Category	Incidents	≤ 500 m	≤ 1 000 m
Tyveri	10	50 %	80 %
Others (Innbrudd, Ulykke, Skadeverk, …)	14	small samples	small samples

4.4 Manual review of street matches

All 127 street-from-text matches from v3 were read by hand. The street name was right in roughly **95 %** of cases. The real problem was long streets: placing «Trondheimsveien 33» at the centroid of an 8 km road gave ±5.7 km. v4 fixed this with exact house numbers, crossings, and limiting long streets to the part near the area named in the area field. The ~6 genuine errors came from streets mentioned as a direction («avkjøringsrampen mot Maridalsveien»), multi-location announcements, and truncated names («Nedre Kalbakkvei» → «Kalbakkvei»); all are handled in v4.

4.5 What cannot be placed (18 %)

- No area field at all (16 incidents).
- Road-only messages: «E6», «Rv4», «E18 Bygdøylokket», «Ring 3 Økern» (~14). Better served by road data (Statens vegvesen DATEX II) later.
- Names spelled differently from the official register, e.g. «Majorstua» vs «Majorstuen», «Bryn», «Alfaset». Fixable with an alias table.
- Places outside Oslo municipality (Lysaker, Hosle in Bærum). Needs per-event municipality lookup.

Caveats

- The «≤ 1 km» figure depends on an assumption: 104 incidents placed on a *tettbebyggelse* name (e.g. Furuset) were given 1 000 m uncertainty. Real-world feedback should calibrate this.
- One quarter, one municipality, one reviewer. Treat numbers as ±5 percentage points.

4.6 Product implications

- Promise neighbourhoods, not street corners: default radius 1 km; street level is a bonus.
- Small radii (< 1 km) MUST only count street-level incidents and MUST say so in the UI.
- Fire and violence, the categories parents care about most, are the best placed (≈ 70–77 % within 1 km).
- Volume is low: ~4.3 incidents/day in all of Oslo, ~2.2/day excluding traffic. A 1 km circle sees only a few per month. Design for quiet weeks.

5. Product strategy

5.1 Positioning

For parents and residents who want to know when something serious happens near the places they care about, stedSafe is a calm alert app that turns the police log into precise, place-based notifications. Unlike the official Politiloggen app, it follows named places instead of whole municipalities, says honestly how precise each incident is, and only interrupts you when it matters.

5.2 Product principles

Principle	What it means in practice
Calm by default	Push only for serious incidents; digest for the rest; «Rolig» is a first-class state; no red sirens, no counters that reward checking.
Honest precision	Every incident carries a precision tier and uncertainty. Never draw a precise pin for an imprecise location.
Place first	The home screen is a list of the user's places with status, not a city-wide feed.
Source-faithful	Show police text as published, with time stamps and NLOD attribution. Never rewrite facts.
Privacy by design	No accounts, minimal data, one-tap delete, no names of children, no tracking SDKs.
No crowd content	No comments, user reports or person descriptions, which avoids the profiling problems seen in Neighbors-style apps.

5.3 Explicit non-goals for the MVP

- No user accounts, social features or comments.
- No live GPS «near me» tracking: places are fixed points, chosen by the user.
- No incident data from other sources (roads, weather, transit) yet; the architecture must allow it.
- No monetisation in v1.

6. MVP scope (product requirements)

Platforms: iOS and Android via Expo. UI language: Norwegian Bokmål. Launch market: Oslo (the pipeline MUST handle any municipality; marketing starts in Oslo).

ID	Feature	Pri.	Acceptance criteria
F1	Create / edit / delete place	P0	Search address or place (Kartverket), or drop a pin; choose template; label ≤ 40 chars; save in < 30 s; delete removes it server-side immediately.
F2	Templates	P0	Selecting a template sets categories and time window (table 6.1). User can override.
F3	Radius + preview	P0	Slider 300–2 000 m, step 100, default 1 000. Shows real count of matching incidents in last 30 days, hidden-traffic count, and the small-radius note when < 1 000 m.
F4	Home overview	P0	One card per place: label, template, radius, window, status («Rolig siste 7 dager» or «N ny»), newest incident with precision badge and distance.
F5	Incident detail	P0	Map with place circle + incident point/uncertainty ring; category; status; precision tier + explanation; «Hvorfor fikk jeg dette?»; all police updates with times; source line.
F6	Push alerts	P0	Only when rules in 6.2 say so; delivered ≤ 2 min after ingestion; tapping opens F5; never twice for the same place + incident.
F7	Weekly digest	P1	Sunday 18:00 Europe/Oslo push + in-app view: per place counts, severe count, imprecise-nearby count; sent even in quiet weeks.
F8	Noise filters	P0	Planned traffic controls never shown; «Trafikk» off by default for Skole/Hjem/Hytte/Jobb.
F9	Settings & privacy	P0	Toggle push; delete all data; privacy policy; source & licence screen.
F10	«Feil plassering?» feedback	P1	One tap on F5 logs event id + place id; used to tune the geocoder.
F11	Family sharing, commuter routes, SEO pages, extra sources	P2	Out of MVP; keep schema extensible.

6.1 Templates

Template	Default categories	Push time window
Skole	Vold, Brann, Savnet, Ro og orden	Weekdays 08:00–16:00
Hjem	Innbrudd, Brann, Vold, Ro og orden	All day
Hytte	Innbrudd, Brann	All day
Jobb	Vold, Brann	Weekdays 07:00–17:00
Annet	All except Trafikk	All day

6.2 Normative event rules

Precision tier from uncertainty u (metres): **gate** if u ≤ 500 · **nabolag** if 500 < u ≤ 1 500 · **bydel** if u > 1 500 or no point.

Match (incident counts for a place) when ALL hold:

1. Incident category is in the place's categories and the incident is not a planned control.

2. distance(place.center, incident.point) ≤ place.radius.
3. Tier is **gate** when radius < 1 000 m; tier is **gate** or **nabolag** when radius ≥ 1 000 m.

Imprecise nearby (digest only, never push): tier **bydel** and the uncertainty circle overlaps the place circle, i.e. distance ≤ radius + u.

Push when the incident matches AND is **severe** AND the first police message is ≤ 30 min old at detection AND the first message time is inside the place's time window AND the place has fewer than 3 pushes today. Otherwise it goes to the digest. Updates to a thread never re-push.

Severe (v1 heuristic, tune in beta): category in {Brann, Voldshendelse, Savnet, Redning, Ulykke} OR text contains any of: `kniv, skudd, skyt, våpen, eksplosjon, evakuer, ran, trusselsituasjon, gissel`. Category Trafikk is severe only if the text contains `alvorlig skadd, omkommet, påkjørt barn`.

7. UX & design specification (direction A)

The chosen direction is **A: soft cards**, with deep teal as the accent. The interactive prototype (private to the founder) lives at `claude.ai/artifact/DGFiKYVJU8We7ut2o3zhLW`. Direction B («Loggbok») is parked; the notification copy rules from B are adopted here.

7.1 Design tokens

Token	Value	Use
color.bg	`#F5F6F3`	App background
color.surface	`#FFFFFF`	Cards
color.border / border.strong	`#E2E5E0 / #C9CFC8`	Card borders / inputs, unselected chips
color.ink / ink2 / muted	`#16201C / #33433B / #5B6660`	Text levels (muted passes 4.5:1 on bg)
color.accent	`#007F82`	Primary buttons, selected chips, Gatenivå badge, map circles
color.accentStrong	`#00676A`	Accent-coloured TEXT on tints and links (5.9:1 on tint)
color.accentTint	`#E3F2F2`	«Hvorfor…» box, digest card, selected category chips
color.severe / severeTint	`#A4501A / #FBEDE3`	Severe category label and «N ny» badge only
color.calmTint	`#F2F4F1`	«Rolig» status rows
radius	`card 18 · inner 12 · button 14 · badge 6 · chip 999`	
spacing	`screen 20 · card padding 16 · stack gap 16 / 12`	
font.body	`Inter 400/500/600/700`	Body, labels
font.display	`Space Grotesk 600/700`	Screen titles, big numbers

7.2 Components

Component	Spec
PlaceCard	Icon tile 40×40 (accentTint bg, accent stroke icon) · label 16/600 · meta 13 muted «Kampen · 1 km · hele døgnet» · optional «N ny» severe badge · status row (calm or incident row).
IncidentRow	Category icon · title 14/600 · PrecisionBadge · «ca. 150 m unna · 08:23». Severe incidents use severeTint background.
PrecisionBadge	**Gatenivå**: filled accent, white text · **Nabolag**: white with 1.5 px accent outline, accent text · **Bydel**: white with 1.5 px dashed grey outline. Differ by shape/fill, not only colour.
TypeChip / CategoryChip	≥ 40 px tall; selected = accent fill (type) or accentTint + check (category); aria-pressed / accessibilityState.selected.
RadiusSlider	300–2 000 m, step 100; live label «1,0 km» / «700 m»; map circle updates live.
InfoNote	Dashed 1 px border, info icon, 13 px text. Used for the small-radius warning.
PrimaryButton	Height 52, radius 14, accent fill, white 16/600 text; pinned to bottom bar.

7.3 Screens

Screen	Content (top to bottom)
Hjem («Dine steder»)	Logo row + settings · title «Dine steder» + «Oppdatert for N minutter siden» · PlaceCards · digest card «Ukesammendrag · uke N» · source line · bottom button «Legg til sted».
Nytt sted	«Avbryt» · search input + selected result · type chips · map with live radius circle · radius slider + label · small-radius InfoNote · preview card («Forhåndsvisning · siste 30 dager», count, hidden traffic) · category chips · «Når» / «Push-varsel» rows · bottom «Lagre sted».
Hendelse	Back «Dine steder» · category + status chips · title · «gate, område · dato tid» · map (place circle dashed + incident ring) · legend · «Hvor presist er dette?» card with badge, explanation, 3-tier legend · «Hvorfor fikk jeg dette?» · timeline «Oppdateringer fra politiet» · actions «Del med familien», «Færre slike varsler» · source line.
Push (OS-rendered)	The OS draws notifications. Only title/body text is ours. See 7.4. The prototype's custom-styled lock screen is illustrative only.

7.4 Notification copy rules

- Title = what + distance + place: «Brann ca. 150 m fra Hjemme».
- Body = where + status + precision: «Hølandsgata, Kampen. Branntilløp i leilighet, nødetatene er på stedet. Presisjon: gate.»
- Never include names, ages, ethnicity or descriptions of people, even if the police text does. Max two lines.
- Digest title: «Uke 46: rolig ved Skolen» or «Uke 46: 4 hendelser rundt stedene dine».

7.5 Accessibility

- Contrast: white on #007F82 = 4.8:1 (pass); #00676A on #E3F2F2 = 5.9:1 (pass). Never put #007F82 text on the tint (4.3:1, fails).
- Touch targets ≥ 44 px. Every icon-only button has an accessibilityLabel.
- Precision and severity are never conveyed by colour alone (badge shape + text label).
- Support Dynamic Type / font scaling; cards grow, nothing truncates critical text.

8. Technical architecture

8.1 Stack

Layer	Choice	Why
Mobile	Expo (React Native, TypeScript), expo-router, react-native-maps, expo-notifications	One codebase for iOS + Android; Expo Push is free
API	FastAPI (Python 3.12), Pydantic v2, SQLAlchemy 2 / asyncpg	Matches founder's stack; geocoder is already Python
Database	PostgreSQL 16 + PostGIS	Geography types, ST_DWithin, GiST indexes
Worker	Separate process (same codebase) with an asyncio loop / APScheduler	Polling, geocoding, matching, push, digest
Hosting	Render (web service + background worker + managed Postgres)	Founder already uses it; low ops
Push	Expo Push API (+ receipts)	Handles APNs/FCM

8.2 Components & data flow

```
Politiloggen API ──(poll 60 s)──► INGESTOR ──► events / event_messages (upsert)
                                      │
                                      ▼
                                  GEOCODER ──► Kartverket address / place-name APIs (cached)
                                      │          point + uncertainty_m + tier + method
                                      ▼
                                CLASSIFIER ──► is_control, severity
                                      │
                                      ▼
                                   MATCHER ──► places (PostGIS ST_DWithin + tier rules)
                                      │
                                      ▼
                                  NOTIFIER ──► Expo Push (immediate) · digest (Sun 18:00)
Mobile app ◄──────── REST API (FastAPI) ◄──────── same database
```

8.3 Politiloggen ingestion

```
GET https://api.politiloggen.politiet.no/messagethreads
    ?take=100&skip=0&sortByEnum=LastMessageOn&timeSpanType=<see note>
Accept: application/json
→ { "messageThreads": [ ... ], "hasMoreResults": true|false }
```

- Field names inside a thread are NOT officially documented. Observed/expected: id, district, municipality, area, category, created/last-message timestamps, active flag, and a list of messages with text and timestamps. The agent MUST fetch a live sample first (the research script has `--sample`), then write a typed adapter with defensive, case-insensitive field lookup (see `pick()` in the reference script).
- Only `timeSpanType=LastQuarter` is verified. Check the API's Swagger for smaller windows; use the smallest that covers the poll interval with margin.
- Sort order is not globally reliable. Each cycle MUST read at least 3 pages (300 threads) and paginate by `skip += len(page)` until `hasMoreResults=false` or the page is empty.
- Upsert by thread id. New thread → geocode → classify → match → notify. Updated thread → store new messages; re-geocode only if area changed or no point yet; never re-push.
- Handle 429/5xx with exponential backoff (1, 2, 4, 8, 16 s). Alert (log at ERROR) if no successful poll for 10 min.

- Map the event's municipality name to a kommunenummer (kommuneinfo API, cached) and geocode within that municipality.
- Attribution: «Kilde: Politiloggen (NLOD 2.0)» on every screen showing police text.

8.4 Database schema

```
CREATE EXTENSION IF NOT EXISTS postgis;

CREATE TABLE events (
  id               TEXT PRIMARY KEY,           -- Politiloggen thread id
  district         TEXT,
  municipality     TEXT,
  kommunenummer    TEXT,
  area             TEXT,
  category         TEXT NOT NULL,
  is_control       BOOLEAN NOT NULL DEFAULT false,
  severity         TEXT NOT NULL CHECK (severity IN ('severe','normal')),
  first_message_at TIMESTAMPTZ NOT NULL,
  last_message_at  TIMESTAMPTZ NOT NULL,
  is_active        BOOLEAN,
  geom             GEOGRAPHY(Point, 4326),
  uncertainty_m    INTEGER,
  precision_tier   TEXT CHECK (precision_tier IN ('gate','nabolag','bydel')),
  geocode_method   TEXT,   -- adresse|kryss|gate_area|gate_tekst|gate_omrade|stedsnavn|kommune
  geocode_match    TEXT,
  raw              JSONB NOT NULL,
  updated_at       TIMESTAMPTZ NOT NULL DEFAULT now()
);
CREATE INDEX events_geom_gix  ON events USING GIST (geom);
CREATE INDEX events_first_idx ON events (first_message_at DESC);

CREATE TABLE event_messages (
  event_id   TEXT REFERENCES events(id) ON DELETE CASCADE,
  seq        INTEGER NOT NULL,
  created_at TIMESTAMPTZ NOT NULL,
  text       TEXT NOT NULL,
  PRIMARY KEY (event_id, seq)
);
```

```
CREATE TABLE devices (
  id           UUID PRIMARY KEY,
  secret_hash  TEXT NOT NULL,          -- bearer secret, hashed
  push_token   TEXT,
  platform     TEXT CHECK (platform IN ('ios','android')),
  tz           TEXT NOT NULL DEFAULT 'Europe/Oslo',
  created_at   TIMESTAMPTZ NOT NULL DEFAULT now(),
  last_seen_at TIMESTAMPTZ
);

CREATE TABLE places (
  id          UUID PRIMARY KEY,
  device_id   UUID NOT NULL REFERENCES devices(id) ON DELETE CASCADE,
  label       TEXT NOT NULL CHECK (char_length(label) <= 40),
  template    TEXT NOT NULL CHECK (template IN ('skole','hjem','hytte','jobb','annet')),
  center      GEOGRAPHY(Point, 4326) NOT NULL,
  radius_m    INTEGER NOT NULL CHECK (radius_m BETWEEN 300 AND 2000),
  categories  TEXT[] NOT NULL,
  time_window JSONB NOT NULL,   -- {"days":[1,2,3,4,5],"from":"08:00","to":"16:00"} | {"all":true}
  push        BOOLEAN NOT NULL DEFAULT true,
  created_at  TIMESTAMPTZ NOT NULL DEFAULT now()
);
CREATE INDEX places_center_gix ON places USING GIST (center);

CREATE TABLE notifications (
  id         BIGSERIAL PRIMARY KEY,
  place_id   UUID REFERENCES places(id) ON DELETE CASCADE,
  event_id   TEXT REFERENCES events(id) ON DELETE CASCADE,
  kind       TEXT NOT NULL CHECK (kind IN ('push','digest')),
  sent_at    TIMESTAMPTZ NOT NULL DEFAULT now(),
  ticket_id  TEXT,
  status     TEXT,
  UNIQUE (place_id, event_id, kind)       -- guarantees no duplicate pushes
);

CREATE TABLE geocode_cache (
  key TEXT PRIMARY KEY, value JSONB, fetched_at TIMESTAMPTZ DEFAULT now()
);
CREATE TABLE placement_feedback (
  event_id TEXT, place_id UUID, created_at TIMESTAMPTZ DEFAULT now()
);
```

8.5 Matching query

```
-- places that should consider a new event :e
SELECT p.*
FROM places p
WHERE :e_category = ANY (p.categories)
  AND NOT :e_is_control
  AND ST_DWithin(p.center, :e_geom, p.radius_m)
  AND (:e_tier = 'gate' OR (:e_tier = 'nabolag' AND p.radius_m >= 1000));
```

8.6 REST API (v1)

Method & path	Purpose
`POST /v1/devices`	Register anonymous device → {device_id, secret}. All other calls: Authorization: Bearer secret.
`PUT /v1/devices/me`	Update push_token, platform, tz.
`DELETE /v1/devices/me`	Delete device, places, notifications (GDPR erase).
`GET /v1/geocode/search?q=`	Proxy to Kartverket address + place-name search (cached, rate-limited).
`GET /v1/preview?lat&lon&radius_m&categories`	Counts for the last 30 days: matched, severe, hidden_traffic, imprecise_nearby. Replaces the prototype's estimate.
`GET/POST /v1/places · PATCH/DELETE /v1/places/{id}`	CRUD for places.
`GET /v1/places/{id}/events?days=30`	Matched + imprecise-nearby events with distance and tier.
`GET /v1/events/{id}`	Event with all messages, point, uncertainty, tier, method.
`POST /v1/events/{id}/feedback`	«Feil plassering?» (body: place_id).
`GET /v1/health`	Liveness + last successful poll time.

8.7 Notifications

- Insert into `notifications` BEFORE sending; the unique constraint prevents duplicates across worker restarts.
- Send via Expo Push in batches of ≤ 100; poll receipts after 15 min; null the token on DeviceNotRegistered.
- Payload data: {event_id, place_id}. Deep link opens the incident detail.
- Digest job: Sundays 18:00 in the device's tz. Per place: matched (7 days), severe, imprecise_nearby. Always send, even if zero («Rolig uke»).

8.8 Privacy & security

- No accounts, no email, no analytics SDK that tracks individuals. Aggregate, anonymous usage counts only.
- Store the minimum: place center, radius, categories, user-chosen label. Advise in the UI: do not use children's names in labels.
- Device secret stored hashed; HTTPS only; rate limit per device.
- One-tap delete (DELETE /v1/devices/me). Retain events (public data) for 13 months; delete inactive devices after 12 months.
- Privacy policy MUST explain that place locations are processed to send alerts (legal basis: consent). v2: evaluate on-device matching so the server never sees places.

8.9 Repository layout

```
stedsafe/
  backend/
    app/  api/  ingest/  geocode/  classify/  match/  notify/  digest/  db/  settings.py
    tests/  fixtures/politiloggen/  fixtures/kartverket/  test_geocode_golden.py
    alembic/
  mobile/   (Expo, TypeScript, expo-router)
    app/(tabs)/index.tsx  app/place/new.tsx  app/event/[id].tsx  components/  theme/tokens.ts
  research/presisjonstest.py      # v4 reference implementation
  docs/stedSafe-spec.pdf
```

9. Geocoding & classification specification

Port the v4 logic of `presisjonstest.py`. Order is significant: the first step that succeeds wins, except for rule 9.4 which can downgrade a street to the area.

9.1 Pipeline

1. **Exact address.** For each street candidate (area field first, then text), look for a house number directly after it (`Trondheimsveien 33`, `Steenstrups gate 1b`). Ignore numbers followed by «etasje», «km», «kmt», «personer», «biler», «år». Look up with `adressenavn, nummer, bokstav, kommunenummer`. Uncertainty 25 m.
2. **Crossing.** Detect «A/B», «A x B», «krysset A og B». Fetch all address points of both streets; take the closest pair; reject if they are > 250 m apart. Uncertainty = max(75, d/2 + 50).
3. **Street.** Candidates from the area field (whole part first, e.g. «Nedre Kalbakkvei»), then from text with the direction filter (9.3). Point = centroid of the street's address points; uncertainty = 90th percentile distance to that centroid (min 50 m).
4. **Long street + area.** If street uncertainty > 1 000 m and the area field resolves to a place name, keep only address points within max(area uncertainty, 800 m) of that place; recompute (min 2 points).
5. **Downgrade.** If the street result is still less precise than the area place name, use the place name.
6. **Place name.** Area parts → place-name API (exact spelling match, prefer area types, see 9.2).
7. **Municipality only.** No point; tier = bydel; never pushed.

9.2 Place-name rules

- Endpoint GET `ws.geonorge.no/stedsnavn/v1/navn?sok=…&knr=…&utkoordsys=4258&fuzzy=false`. The municipality filter is `knr` (not `kommunenummer`); the path is `/navn` (not `/sok`).
- Response: `navn[].stedsnavn[].skrivemåte`, `navn[].navneobjekttype`, `navn[].representasjonspunkt.{øst,nord}`.
- When several objects share the name, prefer by type: administrativ bydel / bydel → tettbebyggelse / tettsteddel → boligfelt → grend → tettsted → industriområde → senter → torg → park → gard → stations → **holdeplass (last)**. «Furuset» in the police log means the area, not the bus stop.
- If the chosen type is «adressenavn», geocode as a street instead.
- Split the area field on «,», «/», «;», « - », « V/ », « ved »; strip road prefixes (E6, Rv. 4, Fv, Ring 3).
- Aliases (police terms that are not official names): «Sentrum», «Oslo sentrum» → 59.9133, 10.7389, 1 200 m; «Oslo S» → 59.9110, 10.7522, 300 m. Add «Majorstua» → «Majorstuen» etc. as found.

Object type	Assumed uncertainty	Object type	Assumed uncertainty
administrativ bydel / bydel	2 500 m	gard (Oslo farm names = neighbourhoods)	700 m
tettsted	2 500 m	industriområde	600 m
tettsteddel	1 200 m	boligfelt	500 m
tettbebyggelse	1 000 m	park	400 m
grend	800 m	stasjon / t-banestasjon	300 m
(default)	1 500 m	holdeplass / skole	200 m
		torg / bru / vegkryss	150 m

9.3 Text parsing rules

- Street suffixes (joined): veien, vegen, veg, vei, gata, gaten, gate, plassen, torget, bakken, stien, bryggen, kaia, ringen, svingen, tunnelen, alléen, hellinga. Separate-word suffixes: gate, gata, vei, veg, plass, torg, allé, bakke, brygge, kai.
- Include up to two capitalised words before a joined name (Økern Torgvei, Nedre Kalbakkvei); generate candidates longest first; drop leading stop-words (I, På, Ved, Mot, Vi, Politiet, Nødetatene, Krysset, …).
- Variants: «-gata/-gaten» → «-gate»; «Hausmannsgate» → «Hausmanns gate».
- Direction filter: skip a street if the preceding 40 chars end with «mot», «retning», «avkjøring(en) mot/til», «kjørte (videre) mot/til», «løp mot», «forsvant (i retning)».
- Planned control: category Trafikk AND text matches trafikkontroll, fartskontroll, laserkontroll, hastighetskontroll, ruskontroll, «aksjon skolevei» or «trafikkorpset har avholdt».

9.4 External-call hygiene

- Cache every successful lookup (including genuine «not found» = HTTP 200 with no hits or 400/404) in `geocode_cache`.
- NEVER cache 429, 5xx, timeouts or connection errors; retry on the next cycle.
- Space calls ≥ 150 ms; exponential backoff on 429.
- Fetch up to 1 000 address points per street (treffPerSide=1000) so crossings and area-limiting work.

9.5 Quality gates (CI)

- All fixtures in Appendix C pass.
- Replay of a recorded Oslo quarter: ≥ 60 % of non-control incidents with uncertainty ≤ 1 000 m and ≥ 25 % ≤ 500 m.
- Manual sample of 25 street-based placements: ≥ 90 % correct street (tracked per release).

10. Go-to-market

- **Launch area:** Oslo, where incident density makes the value obvious. Expand to Akershus municipalities next (Lørenskog, Bærum, Asker).
- **Channels:** FAU (parents' councils) and class groups, neighbourhood Facebook groups, housing co-op boards, local media story about «calm alerts». One parent sharing reaches a whole class.
- **ASO:** «politilogg», «politiloggen varsel», «varsel nabolag», «skole trygghet». Do not use «politi» in the app name or icon.
- **SEO (v2):** public pages per bydel with recent incidents and a calm area summary, linking to the app.
- **Message:** «Rolig oversikt over det som skjer rundt stedene dine.» Never market with fear.

11. Business model & costs

- **v1: free.** Goal is habit and trust.
- **Later premium (don't remove anything free):** more than 3 places, family sharing, 12-month history, route alerts.
- **Later B2B:** FAU/schools, housing co-ops, cabin associations, property managers, estate agents (area history).
- **Running costs:** Render web + worker + managed Postgres on starter tiers (check current pricing); Expo Push, Politiloggen and Kartverket are free. No LLM needed for v1.

12. Risks & mitigations

Risk	Likelihood / impact	Mitigation
Police add radius/place alerts to Politiloggen	Medium / High	Differentiate on templates, digest, precision UX, multi-source; move fast.
API changes or outages (unversioned; outages reported in 2025)	Medium / High	Defensive parsing, contract tests on recorded payloads, health alarms, status banner in app.
Wrong placement causes alarm or misses	Medium / Medium	Precision tiers, conservative push rules, «Feil plassering?» feedback loop.
App increases anxiety	Medium / High	Push only severe, quiet-week messaging, no fear marketing, user-controlled categories.
Privacy breach of place data	Low / High	Minimal data, hashed secrets, deletion, no third-party trackers; v2 on-device matching.
Legal/brand: impersonating police, NLOD attribution	Low / Medium	No police logos or «politi» in name; clear «not affiliated» text; attribution on every screen.
Competitor (Politipuls) improves fast	Medium / Medium	Ship Android + better UX first; build habit via digest.
Solo-founder capacity	High / Medium	Strict MVP scope; reuse existing stack; automate tests.

13. Success metrics

Metric	Target (beta)	Why
Activation: place created in first session	≥ 70 %	Setup is the core value moment
Pushes per active user per week (median)	0.5–2	Calm by design; above 3 signals noise
Push opt-out within 30 days	< 10 %	Trust in alert relevance
Digest open rate	≥ 30 %	Weekly habit

Metric	Target (beta)	Why
D30 retention	≥ 35 %	Product–market fit signal
«Feil plassering?» rate on opened incidents	< 5 %	Geocoder quality
Ingestion lag (publish → push), p95	≤ 2 min	Timeliness

14. Roadmap & milestones

Milestone	Scope	Definition of done
M0 · week 0–1	Parent interviews (3–5) with the prototype · backend spike: ingest + geocode + DB	Live Oslo ingestion running; Appendix C fixtures green; interview notes summarised.
M1 · week 2–4	API, matching, push, Expo screens Hjem / Nytt sted / Hendelse	Internal build (TestFlight + Play internal); push ≤ 2 min end-to-end.
M2 · week 5–6	Digest, settings, delete, privacy policy, store listings	Closed beta with 20–30 Oslo users.
M3 · week 7–10	Iterate on metrics; aliases; Akershus municipalities; feedback loop	Metrics in section 13 met or explained.
Later	Family sharing, commuter routes (Vegvesen DATEX II, Entur), MET weather warnings, SEO pages, B2B	Prioritise from beta learnings.

15. Open questions

1. Exact Politiloggen field names and the allowed values of `timeSpanType` (check Swagger).
2. Is 60-second polling acceptable to the police? No documented rate limit; consider contacting the API owner.
3. Severity heuristics: validate with real beta data and user feedback.
4. Should «bydel»-precision incidents be shown at all outside the digest?
5. Legal review of storing place locations (consent wording, DPIA-light).
6. Name: is «stedSafe» available (trademark, App Store, domain)?
7. Parent interviews: do users understand Gatenivå / Nabolag / Bydel without explanation?

Appendix A · Precision test details

Oslo municipality, Politiloggen LastQuarter window ending 11 Nov 2026, v4 of the research script. 424 threads; 28 planned controls excluded; 396 analysed.

Category	n	≤ 500 m	≤ 1 000 m
Trafikk	195	23 %	57 %
Brann	66	44 %	77 %
Voldshendelse	42	31 %	71 %
Ro og orden	26	27 %	54 %
Andre hendelser	22	32 %	64 %
Savnet	21	5 %	62 %
Tyveri	10	50 %	80 %
Innbrudd	4	25 %	75 %
Ulykke	3	0 %	33 %
Arrangement	2	0 %	0 %
Skadeverk	2	100 %	100 %
Dyr	1	0 %	100 %
Sjø	1	0 %	0 %
Redning	1	100 %	100 %

Place-name type used	n	Radius used
tettbebyggelse	104	1 000 m
alias: Oslo sentrum	25	1 200 m
administrativ bydel	18	2 500 m
gard	12	700 m
holdeplass	4	200 m
industriområde	4	600 m
vik i sjø · tunnel	4 · 4	1 500 m
bru · bruk	3 · 3	150 / 300 m
others (≤ 2 each)	21	per table 9.2

Most frequent unplaceable area values: (empty) 16 · E6 3 · E6 Skulleruddumpa 3 · Rv4 2 · Oslo 2 · Majorstua 2 · Cc Vest 2 · Alfaset 2 · Bryn 2 · E18 Bygdøylokket 2 · Prinsdalen · E6 Inngående Karihaugen · Hosle · Lysaker · Ring 3 Økern Østgående.

Evolution of the test: v1 had a broken place-name call (0 % place names). v2 fixed the endpoint (`/navn`, `knr`). v3 preferred area types over bus stops and stopped caching failed calls. v4 added house numbers, crossings, area-limited long streets, the direction filter and the control filter.

Appendix B · Category mapping & UI strings

App category (UI)	Politiloggen categories
Vold	Voldshendelse
Brann	Brann

App category (UI)	Politiloggen categories
Savnet	Savnet
Ro og orden	Ro og orden
Innbrudd	Innbrudd
Tyveri	Tyveri
Trafikk	Trafikk (minus planned controls)
Andre (off by default)	Ulykke, Redning, Skadeverk, Sjø, Dyr, Vær, Arrangement, Andre hendelser

Key	Norwegian string
`home.title`	Dine steder
`home.updated`	Oppdatert for {n} minutter siden
`home.add`	Legg til sted
`place.calm`	Rolig siste 7 dager
`place.new`	{n} ny
`new.title`	Nytt sted
`new.search`	Søk etter adresse, skole eller sted
`new.type`	Type sted
`new.radius`	Område rundt stedet
`new.smallRadius`	I et så lite område teller bare hendelser der politiet oppgir gate eller adresse. Det er omtrent 1 av 4 hendelser.
`new.preview`	Forhåndsvisning · siste 30 dager
`new.categories`	Varsle meg om
`new.when / new.push`	Når / Push-varsel · Bare alvorlige
`new.save / cancel`	Lagre sted / Avbryt
`event.precision`	Hvor presist er dette?
`event.why`	Hvorfor fikk jeg dette?
`event.updates`	Oppdateringer fra politiet
`event.share / fewer`	Del med familien / Færre slike varsler
`tier.gate / nabolag / bydel`	Gatenivå / Nabolag / Bydel
`source`	Kilde: Politiloggen (NLOD 2.0). stedSafe er ikke tilknyttet politiet.

Appendix C · Geocoder test fixtures

Each fixture: area field, message text (excerpt) → expected result. Use recorded Kartverket responses so tests run offline.

#	Input	Expected
1	text «Nødetatene er på vei til Trondheimsveien 33 etter melding om brann i 3 etasje»	method adresse · Trondheimsveien 33 · tier gate (not house number 3)
2	area «Rosenhoff», text «…sammenstøt … i Trondheimsveien»	method gate_omrade · points near Rosenhoff only · u < 1 000 m

#	Input	Expected
3	text «I krysset Louises gate/Ullevålsveien har det vært en trafikkulykke»	method kryss · ≈ 75–150 m
4	area «Tåsentunnelen», text «…avkjøringsrampen mot Maridalsveien»	NOT Maridalsveien (direction filter)
5	text «…hendelsen er i Økern Torgvei 1»	method adresse · Økern torgvei 1
6	area «Nedre Kalbakkvei»	street Nedre Kalbakkvei (not Kalbakkvei)
7	area «Mosseveien V/ Ulvøya»	parts [Mosseveien, Ulvøya]; long street limited to Ulvøya
8	category Trafikk, text «Politiet har hatt trafikkontroll i Holtegata»	is_control = true
9	area «E6 Furuset»	place name Furuset (prefix stripped)
10	area «Furuset» with API hits {Holdeplass, Tettbebyggelse}	type tettbebyggelse · 1 000 m
11	area «Sentrum»	alias Oslo sentrum · 1 200 m
12	text «Holtegata hvor fartsgrensen er 30 km/t»	no house number
13	text «krysset Hausmannsgate/Storgata»	crossing of «Hausmanns gate» and «Storgata»
14	area «Bjørndal», text «…sør for Enebakkveien»	place name Bjørndal (street less precise → downgrade)
15	Kartverket returns HTTP 429	result None and NOT cached; retried next cycle

Appendix D · Glossary

Term	Meaning
Politiloggen	The Norwegian police's public operations log; replaced X/Twitter as the main live channel.
Operasjonssentral	Police operations centre that publishes to Politiloggen.
Kommune / kommunenummer	Municipality / its 4-digit code (Oslo = 0301).
Bydel	City district (Oslo has 15).
Tettbebyggelse	Built-up area name in the place-name register (e.g. Furuset); used here as «neighbourhood».
SSR	Sentralt stedsnavnregister: Kartverket's place-name register.
NLOD 2.0	Norwegian Licence for Open Government Data; requires attribution.
FAU	Foreldrenes arbeidsutvalg: parents' council at a school.
Gatenivå / Nabolag / Bydel	The app's precision tiers (≤ 500 m / ≤ 1 500 m / coarser).

Appendix E · Sources

- Politiet: Politiloggen and the Politiloggen app: politiet.no/politiloggen
- Politiloggen app, App Store (Norway): apps.apple.com/no/app/politiloggen/id6504481490
- Politipuls, API guide: politipuls.no/politiloggen/ · about: politipuls.no/om · support: politipuls.no/support
- Politipuls, App Store: apps.apple.com/no/app/politipuls-police-norway/id6760539115
- Kartverket place-name API user guide: kartverket.no/en/api-and-data/stedsnavndata/brukarrettleiing-stadnamn-api

- Kartverket place-name API listing: data.norge.no (Åpent stedsnavn API fra Kartverket) · ws.geonorge.no/stedsnavn/v1/openapi.json
- Kartverket address API: ws.geonorge.no/adresser/v1
- Allemannsdata MCP servers: allemannsdata.com
- Politikontroller, Google Play listing (500k+ drivers)
- Poliskollen, App Store listing (id6757537288)
- Brottsplatskartan, Home Assistant integration: home-assistant.io/integrations/brottsplatskartan
- Nabohjælp: App Store listing; Ritzau press release on ~290,000 users
- Mother Jones: «It Creates a Culture of Fear»: how crime tracking apps incite unnecessary panic (2019)
- Statens vegvesen DATEX II · MET MetAlerts 2.0 · Entur real-time API (future sources)
- Internal: presisjonstest.py v1–v4 output and manual review, 11 Nov 2026